Magnetism

Grade 9 – Semester 2

Magnet is an object that creates a magnetic field which we can't see. A magnet is able to attract magnetic materials such as iron, nickel, cobalt and steel. However, magnets can't attract non magnetic materials which are plastic, wood, paper, aluminium copper, lead, brass and rubber. All magnets have 2 sides, which are N (north pole) and S (south pole)

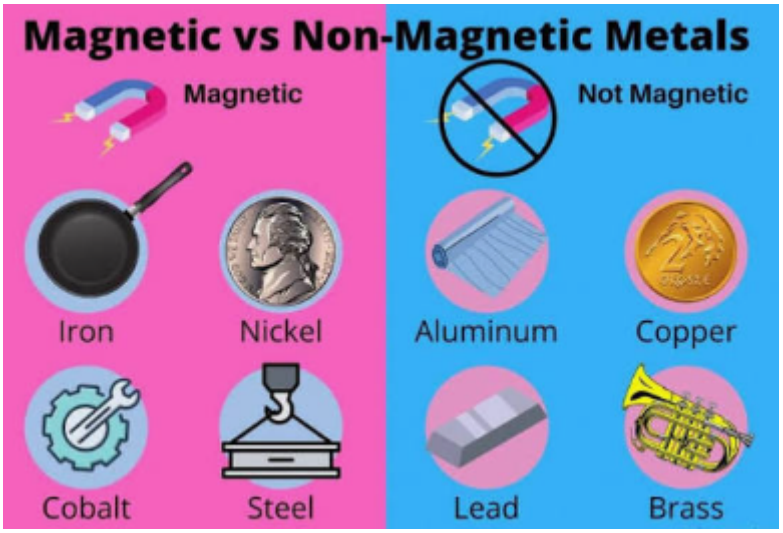

Figure 1 Magnetic Vs Non-Magnetic Metals

Magnetic Poles

In a magnetic pole, there are 2 fields. N (north) and S (south). If N and N or S and S meet, it'll repel each other. Meanwhile, if N and S meet, the 2 magnets will be attracted to each other.

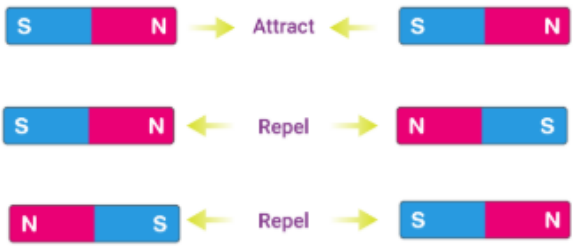

Figure 2 Magnetic Poles

Magnetic Field

Magnetic field is a volume of space that can detect magnetism. A magnetic field can't be seen, however with the help of iron fillings, it can show us how the magnetic field looks and discover its strength and direction.

We can know how strong a magnetic field is by seeing how close the field lines are to each other. A weak magnetic field will have a huge gap between each line.

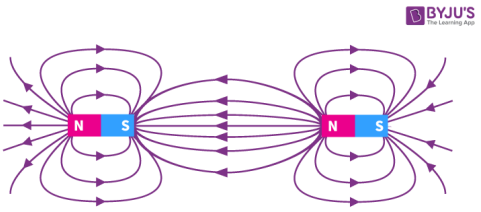

Figure 3 Magnetic field Attracted

Induced Magnetism

In magnets, there are 2 types of magnetic material which are soft material and hard magnet.

A soft magnetic material means that after its being magnetized, it is temporary and the magnetism will disappear if the permanent magnet is removed. Examples of these materials are iron and nickel. Meanwhile, when a hard material is magnetized, it is permanent and can't be removed. Examples of hard magnetic materials are steel.

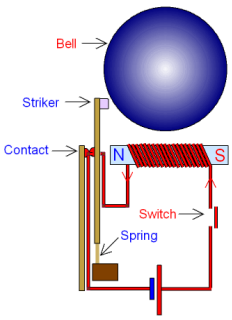

Figure 4 Example of Soft Magnetic Material, Traditional School Bell

Figure 5 Example of Hard Magnetic Material, Bar Magnet

Magnets

In order to make magnets, we will be needing a magnetic material (it could be a soft or hard magnetic material). Grab the magnetic materials and rub a magnet on it. It'll then become magnetised.

To remove its magnetism (only for soft materials), put heat below it and the magnetism will soon shake and leave the object. This process is called demagnetization.

Electromagnetism

Electromagnetism is when a current in a wire in a magnetic field will later be created around it. The field around it will later form a circular shape.

To figure out the direction of the magnetic field, we use a technique called the right-hand grip rule. Our thumbs are used to measure the direction of the current while the rest of our fingers measure the direction of the magnetic field.

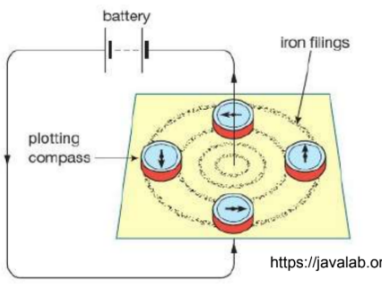

Figure 6 Electromagnet

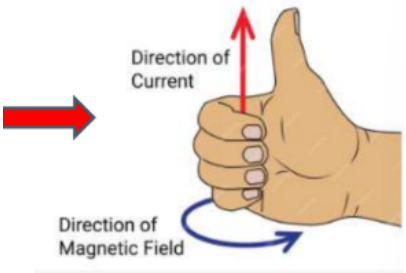

Figure 7 Right-Hand Grip Rule